

Experiment 01:

7. Laboratory Exercise

a. Answer the following.

Q1. Define Web and compare with the term Internet.

Ans:

- **Definition of Web**

The **World Wide Web (WWW)**, commonly called the **Web**, is a system of interconnected web pages and resources that are accessed over the Internet using web browsers. It uses protocols such as **HTTP** and **HTTPS** to deliver text, images, videos, and other multimedia content. The Web enables users to access information through hyperlinks, making navigation between documents easy and interactive.

- **Comparison between Web and Internet:**

Parameter	Web (World Wide Web)	Internet
Definition	A collection of interconnected websites and web pages.	A global network of interconnected computer networks.
Purpose	Provides access to web pages and online content.	Enables communication and data transfer between devices worldwide.
Dependency	Depends on the Internet to function.	Can exist without the Web.
Protocols	HTTP, HTTPS	TCP/IP, FTP, SMTP, DNS, HTTP, etc.
Access Method	Through web browsers like Chrome, Firefox, Edge.	Through browsers, email clients, FTP software, messaging apps, etc.
Services	Websites, blogs, online shopping, web applications.	Email, file transfer, VoIP, web services, cloud computing, streaming, etc.

Q2. What is a Web-Based Application?**Ans:**

A **Web-Based Application (Web App)** is software that runs on a web server and is accessed through a web browser over the Internet or an intranet. Unlike traditional desktop applications, web applications do not require installation on the user's device. They allow users to perform various tasks such as communication, online shopping, banking, education, and data management from any device with internet connectivity.

• Characteristics

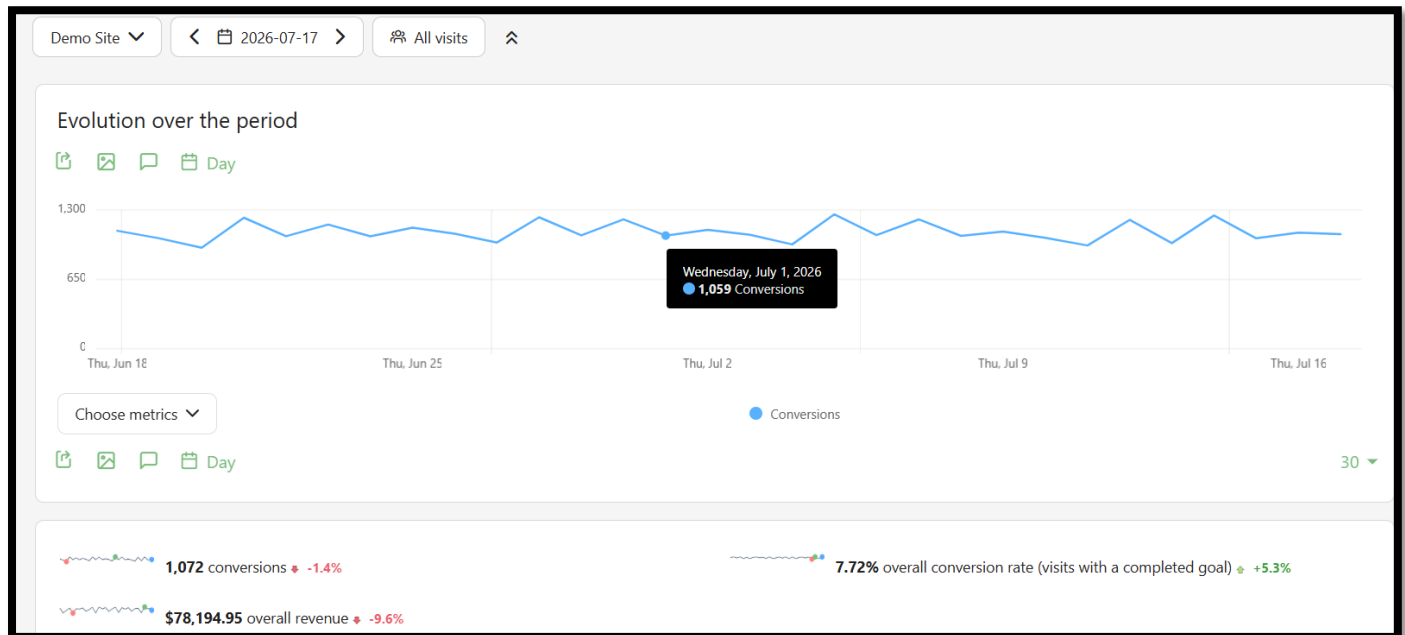
- Accessible through a web browser.
- Platform-independent and works on multiple operating systems.
- No installation required on the client device.
- Data is stored and processed on a remote server.
- Easy to maintain and update since changes are made on the server.
- Supports multiple users simultaneously.

• Examples

- Gmail
- Google Docs
- Facebook
- Amazon
- Microsoft Office 365

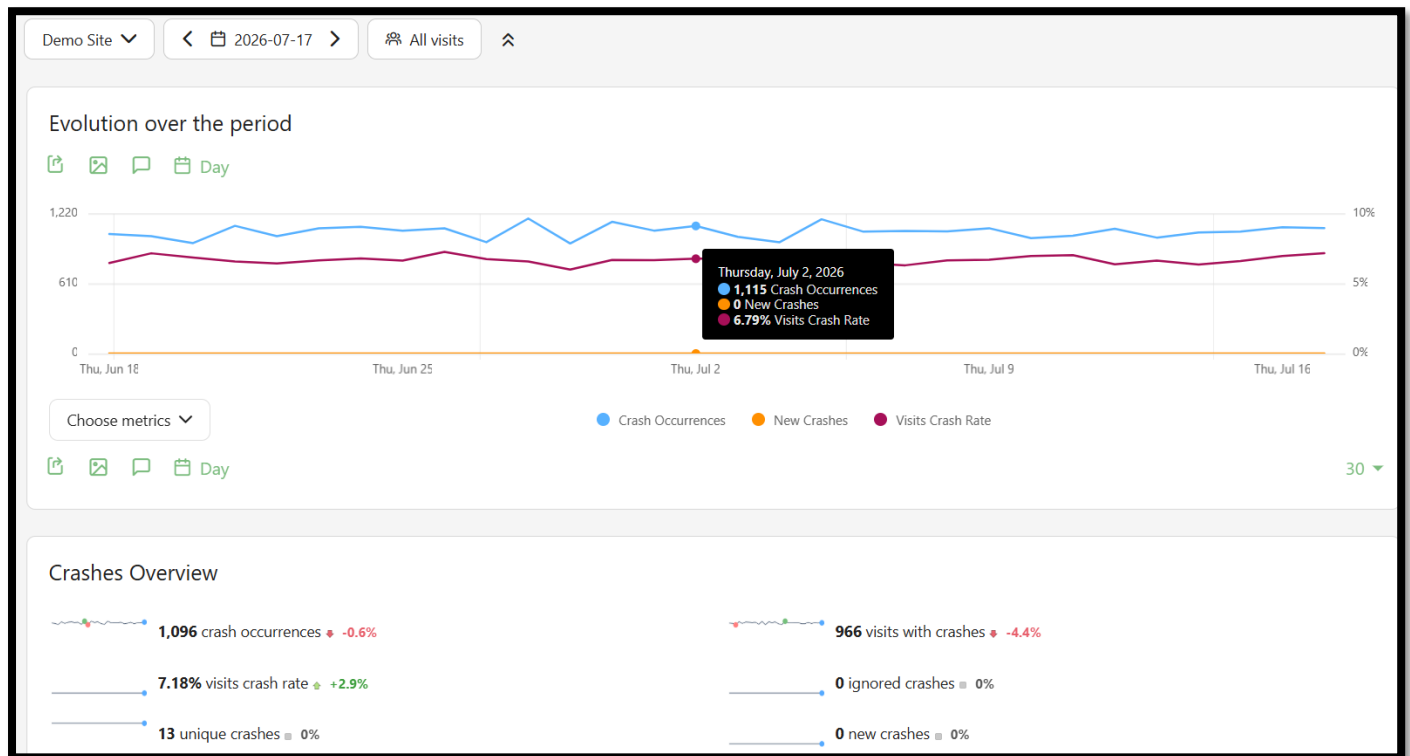
Q3. Compare Google Analytics with Countly, Plausible, and Matomo.**Ans:**

Sr. No.	Parameter	Google Analytics	Countly	Plausible	Matomo
1	Session	Tracks user sessions with detailed engagement metrics.	Tracks sessions with real-time analytics and user behaviour.	Tracks sessions using privacy-friendly methods.	Tracks sessions with complete ownership of collected data.
2	Total Visitors	Provides total users and active users with advanced reports.	Displays total visitors in real time with user segmentation.	Displays total visitors through a simple dashboard.	Provides total visitors with customizable reports.
3	Returning Users/Visitors	Identifies new and returning visitors with detailed audience analysis.	Measures returning users using user profiles and retention analytics.	Shows returning visitors while preserving user privacy.	Provides detailed reports on new and returning visitors.
4	Average Request Received	Tracks page views, events, and server requests with extensive metrics.	Measures API requests, events, and application usage.	Focuses mainly on page views and event tracking with minimal data collection.	Tracks page requests, events, downloads, and user interactions.
5	Time Spent	Measures average engagement time and session duration.	Measures session duration and user engagement in real time.	Displays average visit duration with lightweight analytics.	Measures average visit duration and detailed engagement statistics.

b. Attach screenshots:**• Web Evolution:****Observation 1: Conversions Dashboard of Evolution**

- ❖ The dashboard displays the **Conversions Analytics** section, which tracks the number of successful user conversions over a selected period.
- ❖ The "**Evolution over the period**" line graph visualizes daily conversion trends, helping identify fluctuations in user engagement.
- ❖ On **1 July 2026**, the dashboard records **1,059 conversions**, indicating consistent website activity.
- ❖ The graph shows relatively stable conversion values with minor peaks and dips, reflecting steady user interactions.
- ❖ Key performance metrics displayed include:
 - **Total Conversions:** 1,072
 - **Overall Conversion Rate:** 7.72%
 - **Overall Revenue:** \$78,194.95
- ❖ Percentage indicators compare current performance with the previous period, enabling trend analysis.
- ❖ Filters such as **Site Selection**, **Date Range**, and **Visitor Segmentation** allow customized reporting and analysis.

- ❖ The dashboard helps evaluate the effectiveness of marketing campaigns, user engagement, and business performance.
- ❖ Overall, the dashboard provides actionable insights for improving website conversions and achieving business objectives.

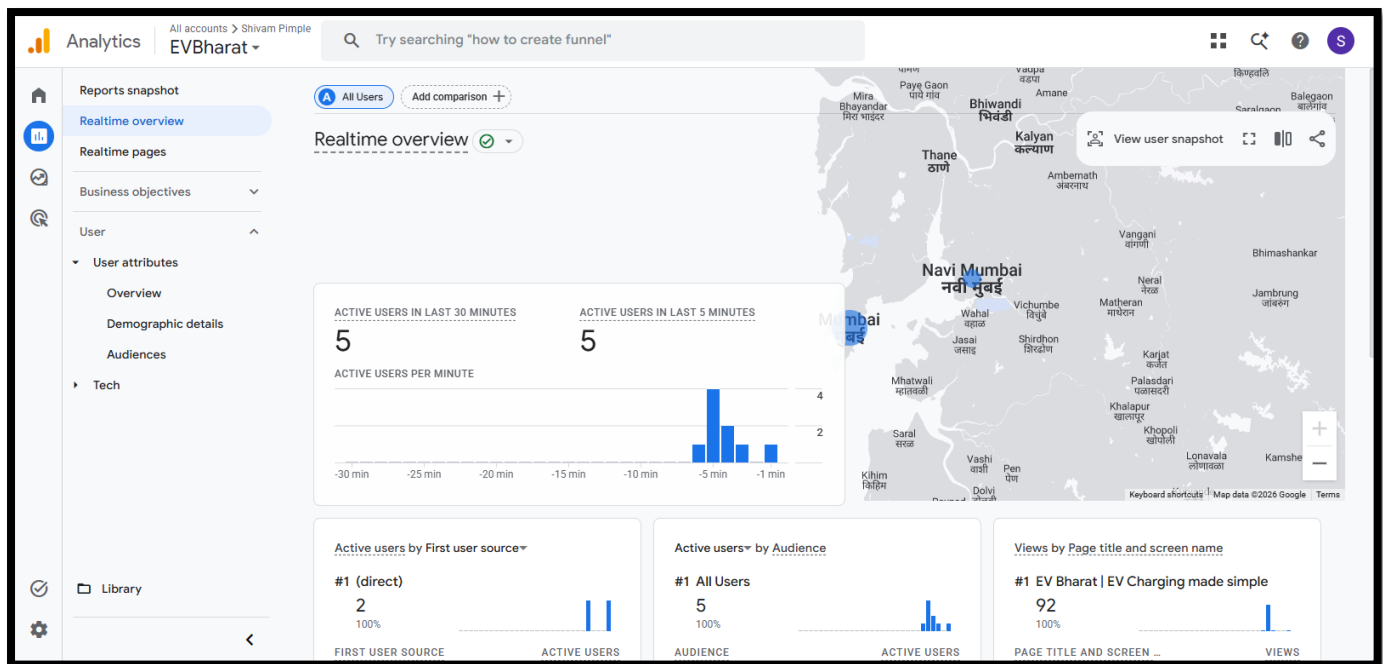


Observation 2: Crashes Analytics Dashboard of Evolution

- ❖ The dashboard displays the **Crash Analytics** section, which monitors application stability and crash-related events.
- ❖ The "**Evolution over the period**" graph tracks three important metrics:
 - Crash Occurrences
 - New Crashes
 - Visits Crash Rate
- ❖ On **2 July 2026**, the dashboard reports:
 - **1,115 Crash Occurrences**
 - **0 New Crashes**
 - **6.79% Visits Crash Rate**

- ❖ The graph helps developers identify crash trends and assess application reliability over time.
- ❖ The **Crashes Overview** section summarizes important statistics:
 - **Total Crash Occurrences:** 1,096
 - **Visits with Crashes:** 966
 - **Crash Rate:** 7.18%
 - **Unique Crashes:** 13
 - **Ignored Crashes:** 0
 - **New Crashes:** 0
- ❖ Percentage comparisons indicate whether crash frequency has increased or decreased compared to the previous period.
- ❖ The dashboard enables developers to detect recurring issues, monitor software quality, and improve application stability.
- ❖ Overall, this analytics view supports proactive debugging, better system reliability, and an enhanced user experience.

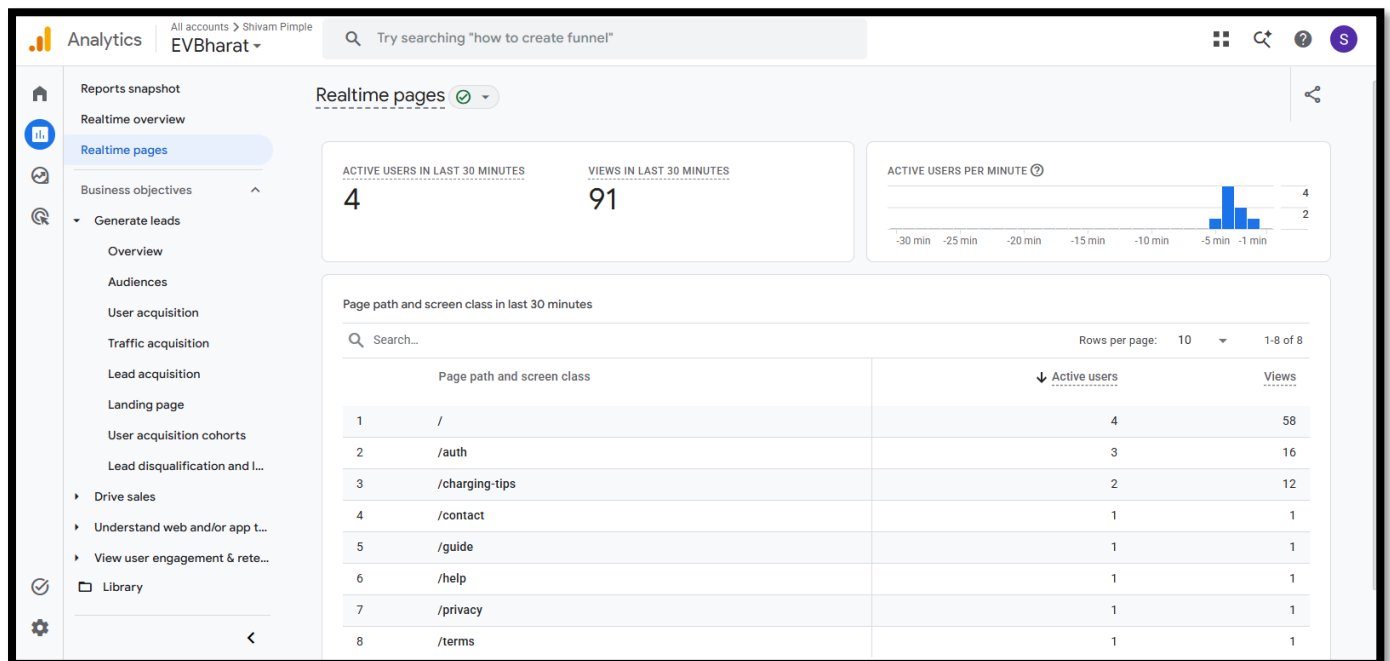
• Google Analytics home page:



Observation 1: Realtime Overview

- ❖ The screenshot displays the **Realtime Overview** dashboard of **Google Analytics 4 (GA4)**, which provides live insights into user activity on the website.
- ❖ The dashboard reports:
 - **5 Active Users** in the last **30 minutes**.
 - **5 Active Users** in the last **5 minutes**, indicating that all current users are actively interacting with the website.
- ❖ The **Active Users per Minute** bar chart visualizes user activity over the previous 30 minutes, making it easy to identify traffic spikes.
- ❖ A **geographical map** highlights the real-time locations of active users, helping administrators monitor visitor distribution across different regions.
- ❖ The **Active Users by First User Source** panel identifies the acquisition source of visitors. In this snapshot, users have primarily arrived through the **Direct** channel.
- ❖ The **Active Users by Audience** section categorizes visitors into audience segments, enabling targeted analysis of different user groups.
- ❖ The **Views by Page Title and Screen Name** panel indicates that the page "**EV Bharat | EV Charging made simple**" has received **92 views**, making it the most actively viewed page.

- ❖ Filters such as **User Segments**, **Comparisons**, and **Date Selection** allow administrators to customize the displayed analytics.
- ❖ Overall, the Realtime Overview dashboard enables website administrators to monitor live traffic, analyse visitor behaviour, evaluate traffic sources, and make timely decisions based on current website activity.

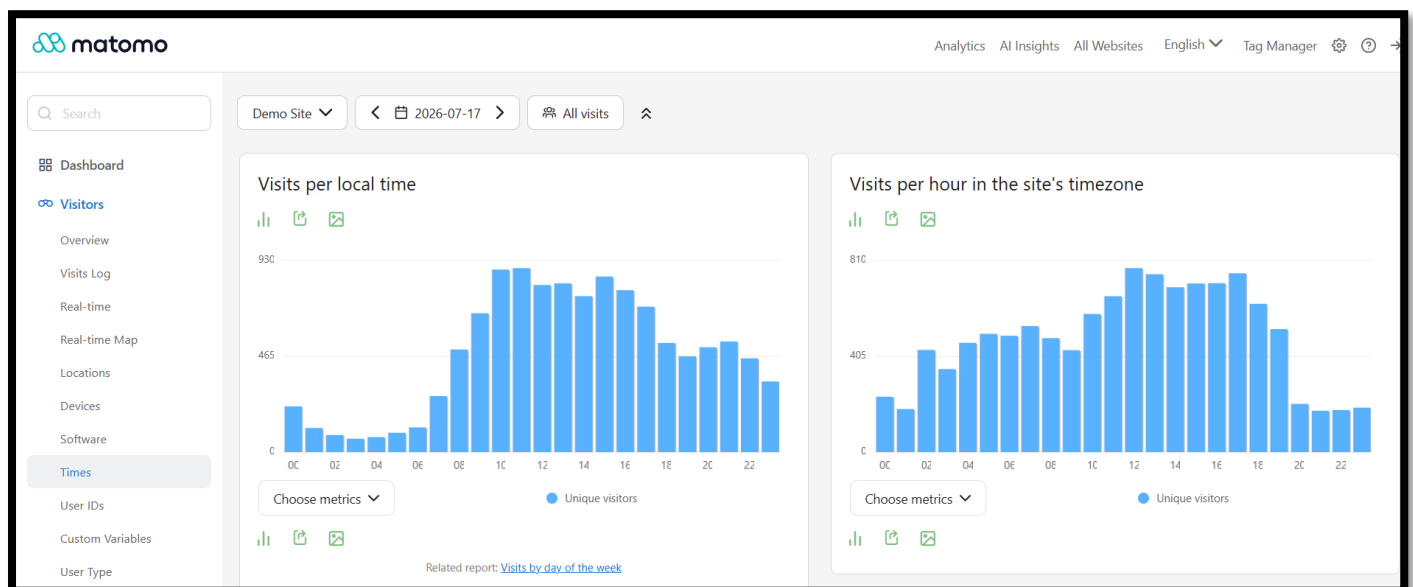


Observation 2: Realtime Pages

- ❖ The screenshot shows the **Realtime Pages** report in **Google Analytics 4**, which provides live statistics for individual web pages.
- ❖ The dashboard displays:
 - **4 Active Users** in the last **30 minutes**.
 - **91 Total Page Views** during the same period.
- ❖ The **Active Users per Minute** graph illustrates user activity trends over the past 30 minutes, helping identify periods of increased engagement.
- ❖ The **Page Path and Screen Class** table lists all pages currently receiving traffic, along with the number of **Active Users** and **Views** for each page.
- ❖ The homepage (/) records the highest engagement with:
 - **4 Active Users**
 - **58 Views**

- ❖ Other pages such as **/auth**, **/charging-tips**, **/contact**, **/guide**, **/help**, **/privacy**, and **/terms** also receive user visits, indicating navigation across multiple sections of the website.
- ❖ The table enables developers and website administrators to identify the most frequently visited pages and understand user navigation patterns.
- ❖ This report is particularly useful for monitoring the immediate impact of newly published content, marketing campaigns, or promotional events.
- ❖ Overall, the Realtime Pages dashboard provides valuable insights into page-level user engagement, helping optimize website content, improve navigation, and enhance the overall user experience

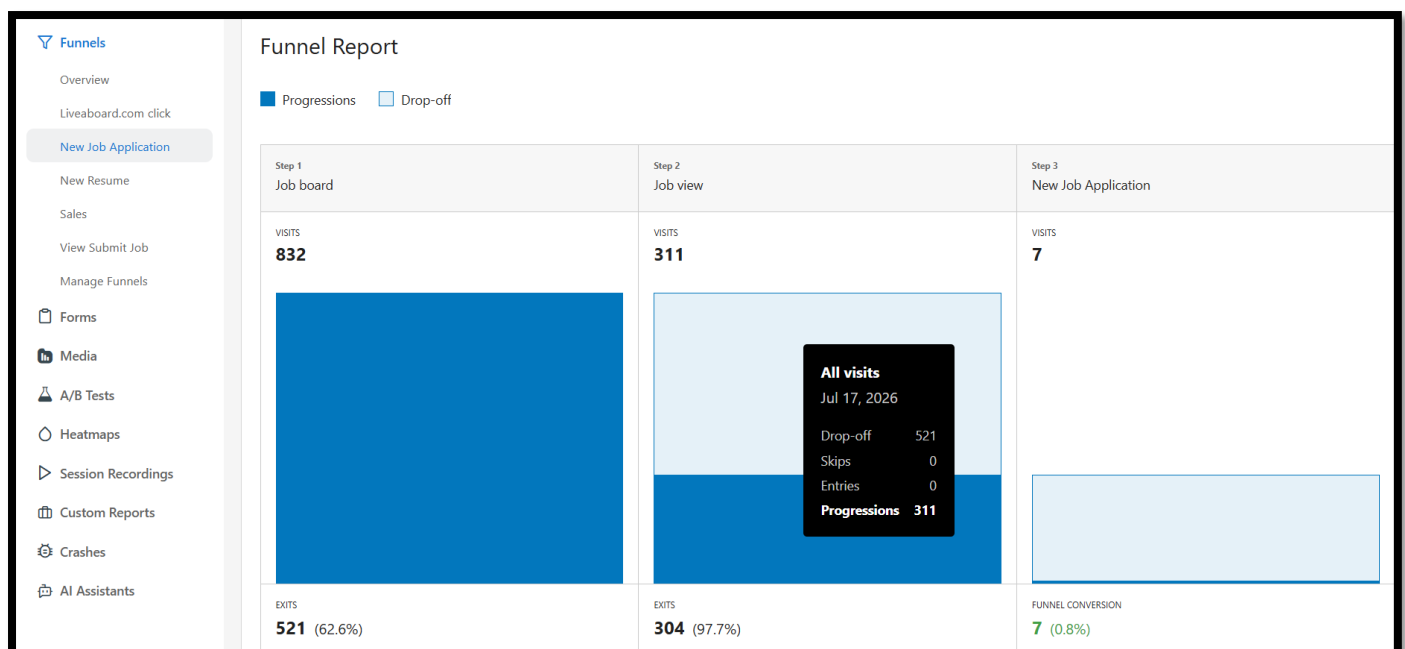
• Matomo user interface:



Observation 1: Visitor Time Analysis

- ❖ The screenshot displays the **Times** report under the **Visitors** section of the **Matomo Analytics** dashboard.
- ❖ It contains two bar charts:
 - **Visits per Local Time**
 - **Visits per Hour in the Site's Time Zone**
- ❖ The **Visits per Local Time** chart shows visitor activity according to each user's local time, helping identify when users are most active across different geographical regions.
- ❖ The **Visits per Hour** chart represents website traffic based on the website's configured time zone, making it easier to schedule content publishing and server maintenance.

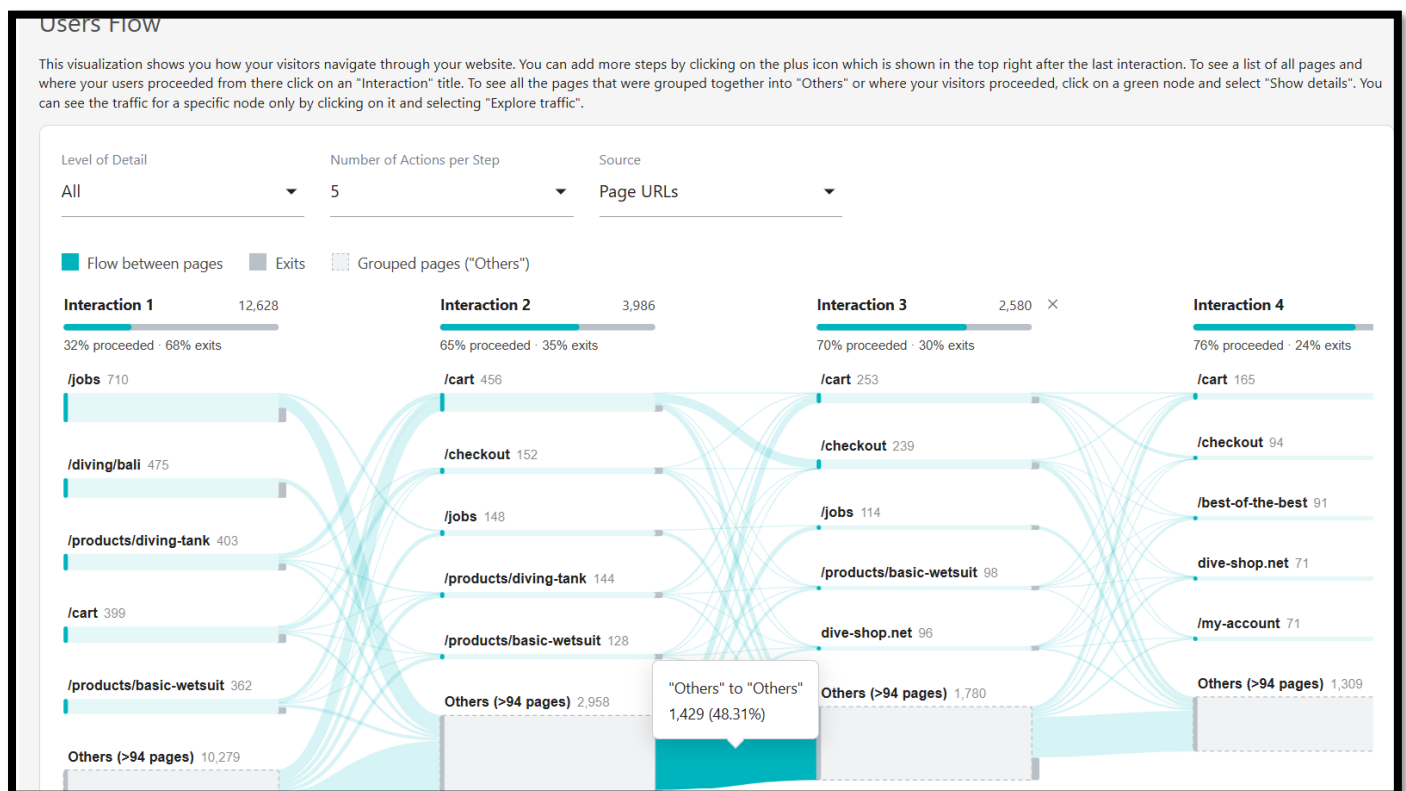
- ❖ Visitor activity gradually increases during the morning hours, reaches its peak around **midday to afternoon**, and declines during late evening and night.
- ❖ The dashboard provides options to customize displayed metrics using the **Choose Metrics** dropdown.
- ❖ Date filters, website selection, and visitor segmentation enable detailed traffic analysis for different time periods.
- ❖ This report helps website administrators determine the optimal time for content releases, promotional campaigns, and system maintenance.
- ❖ Overall, the dashboard provides valuable insights into user activity patterns, enabling organizations to optimize website performance and improve user engagement.



Observation 2: Funnel Report

- ❖ The screenshot shows the **Funnel Report**, which visualizes how users progress through different stages of a predefined conversion process.
- ❖ The funnel consists of three sequential stages:
 - **Step 1: Job Board**
 - **Step 2: Job View**
 - **Step 3: New Job Application**
- ❖ The report records:
 - **832 Visits** at the Job Board stage.

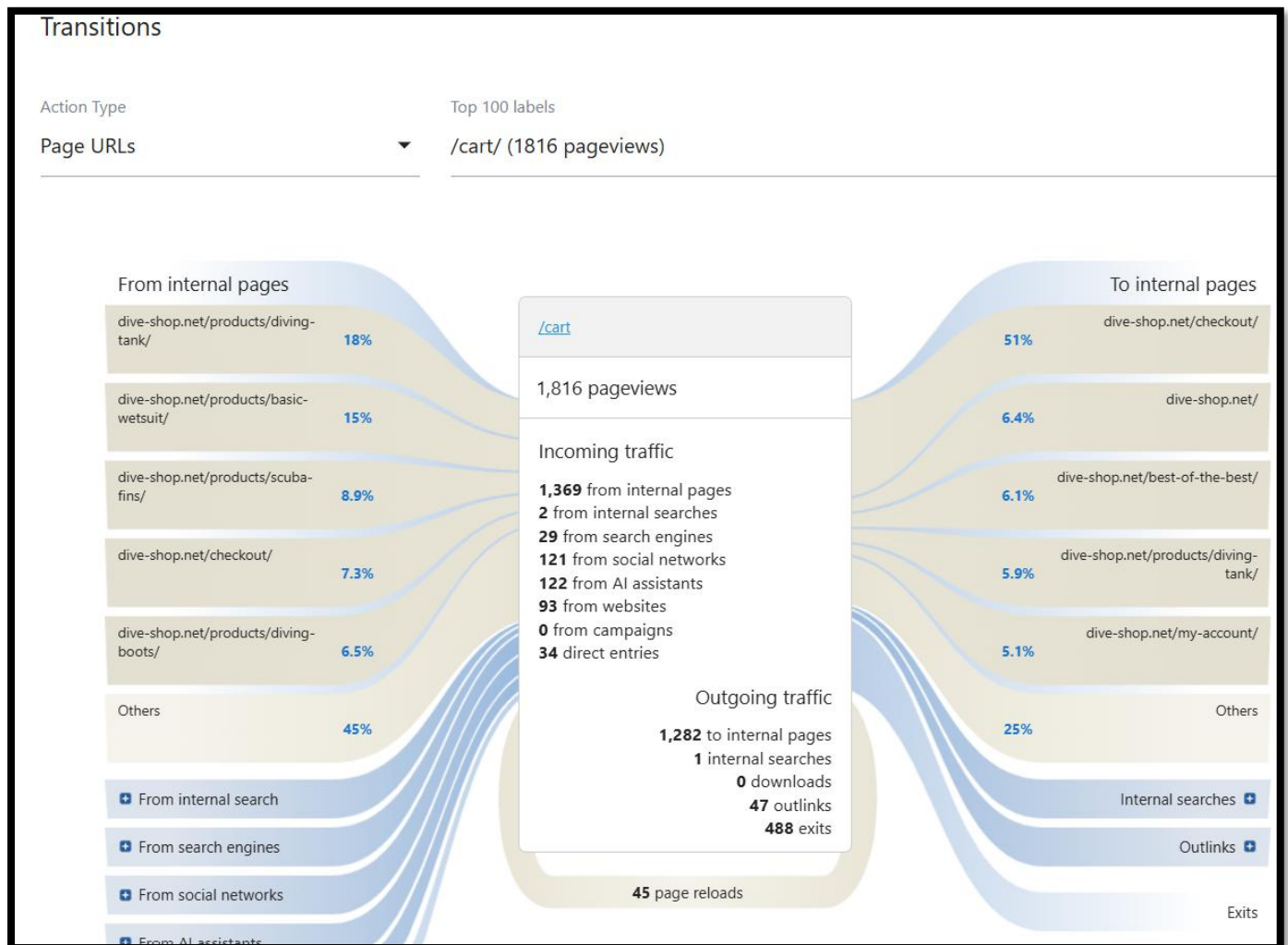
- **311 Visits** reaching the Job View stage.
 - **7 Visits** completing the New Job Application stage.
- ❖ The visualization distinguishes between **Progressions** (users moving to the next step) and **Drop-offs** (users leaving the process before completion).
 - ❖ A significant number of users exit after the first and second stages, indicating potential usability or engagement issues within the application process.
 - ❖ The overall **Funnel Conversion Rate** is **0.8%**, highlighting opportunities to optimize the user journey and improve conversions.
 - ❖ This report helps organizations identify bottlenecks in customer workflows and evaluate the effectiveness of their conversion strategies.
 - ❖ Overall, the Funnel Report is an effective tool for analysing user behaviour and improving website or application conversion performance.



Observation 3: Users Flow

- ❖ The screenshot presents the **Users Flow** visualization, which illustrates how visitors navigate between different pages of the website.
- ❖ The report displays user interactions across multiple navigation steps, showing both successful page transitions and user exits.

- ❖ Each interaction stage contains:
 - The number of visitors.
 - The percentage of users proceeding to the next step.
 - The percentage of users exiting the website.
- ❖ The visualization uses **flow diagrams** to represent movement between popular pages such as **/jobs**, **/cart**, **/checkout**, and various product pages.
- ❖ Grouped pages labelled "**Others**" combine less frequently visited pages, improving readability of the report.
- ❖ Administrators can customize the visualization by changing:
 - Level of Detail
 - Number of Actions per Step
 - Source (e.g., Page URLs)
- ❖ The report enables analysts to understand common navigation paths, identify abandoned journeys, and detect pages where users frequently exit.
- ❖ Overall, the Users Flow report provides comprehensive insights into visitor navigation behaviour, supporting website optimization and enhanced user experience.



Observation 4: Matomo – Page Transitions

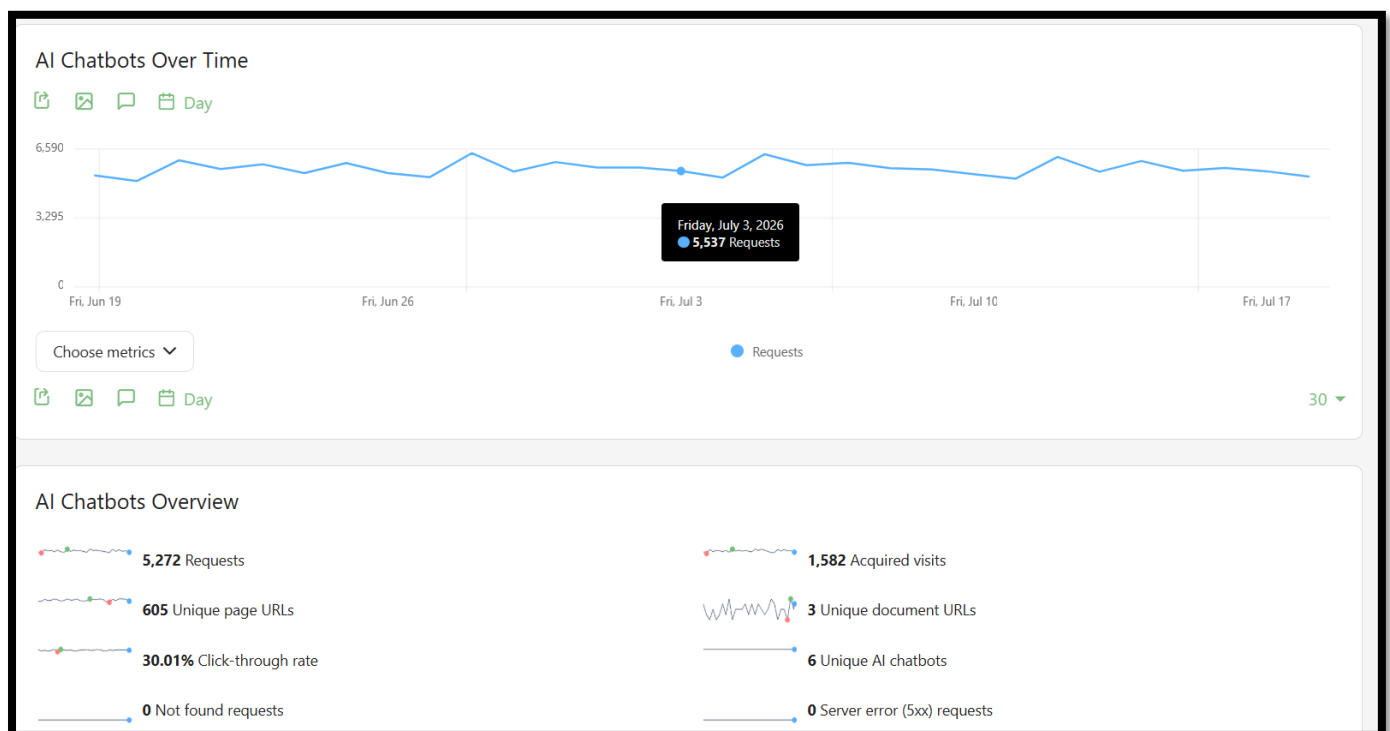
- ❖ The screenshot displays the **Transitions** report, which analyses how visitors arrive at and leave a specific webpage.
- ❖ The selected page is **/cart**, which has recorded **1,816 page views**.
- ❖ The report categorizes **Incoming Traffic** from multiple sources, including:
 - Internal website pages
 - Internal searches
 - Search engines
 - Social networks
 - AI assistants
 - External websites
 - Direct entries

- ❖ It also presents **Outgoing Traffic**, showing where visitors navigate after viewing the selected page.
- ❖ The largest proportion of outgoing traffic moves to the **checkout page (51%)**, indicating that many visitors continue the purchasing process.
- ❖ Additional navigation paths include product pages, account pages, internal searches, outlinks, and exits.
- ❖ The Sankey-style visualization clearly represents visitor movement between pages, making navigation analysis more intuitive.
- ❖ This report helps identify popular navigation paths, optimize website structure, improve internal linking, and reduce user drop-offs.
- ❖ Overall, the Transitions report provides detailed page-level navigation analytics, enabling data-driven improvements in website usability and conversion optimization.

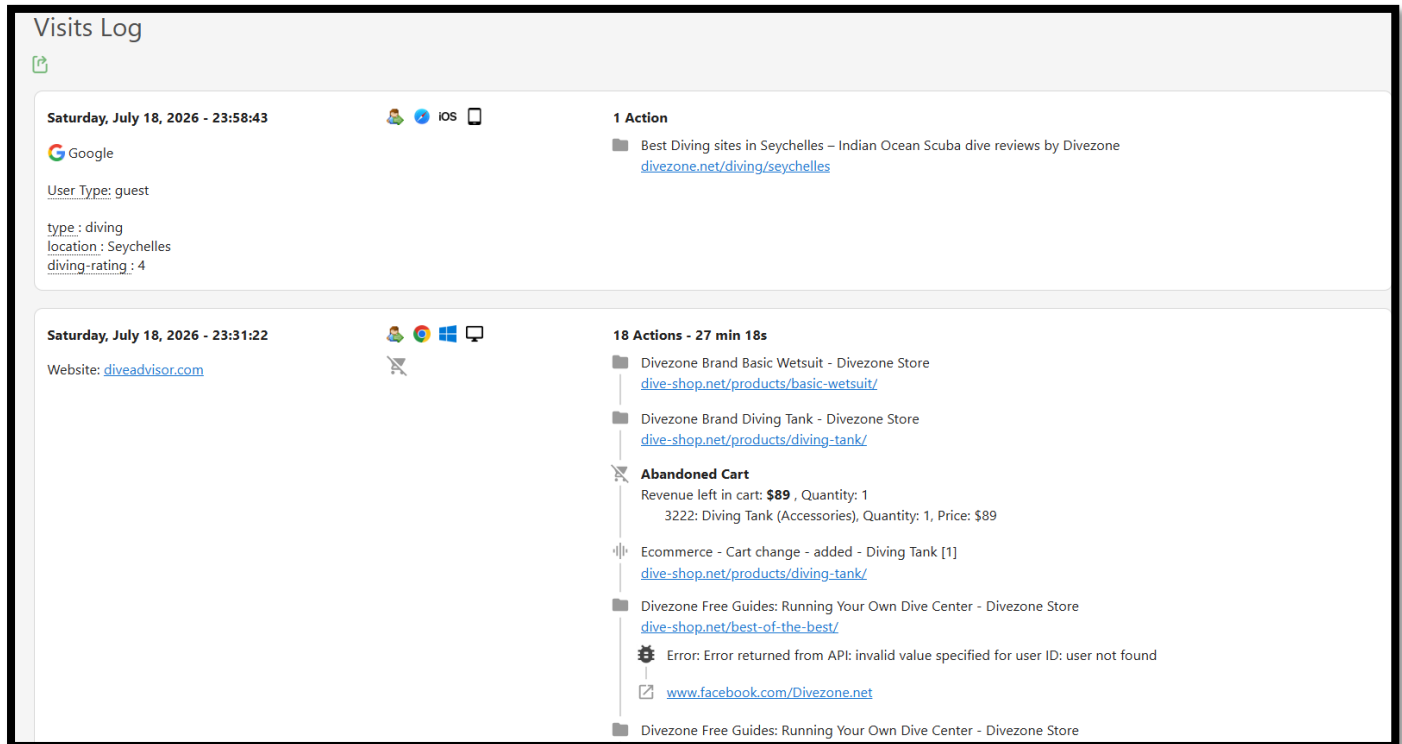
Post Experiments Exercise:

- Screenshots of new feature in Matomo

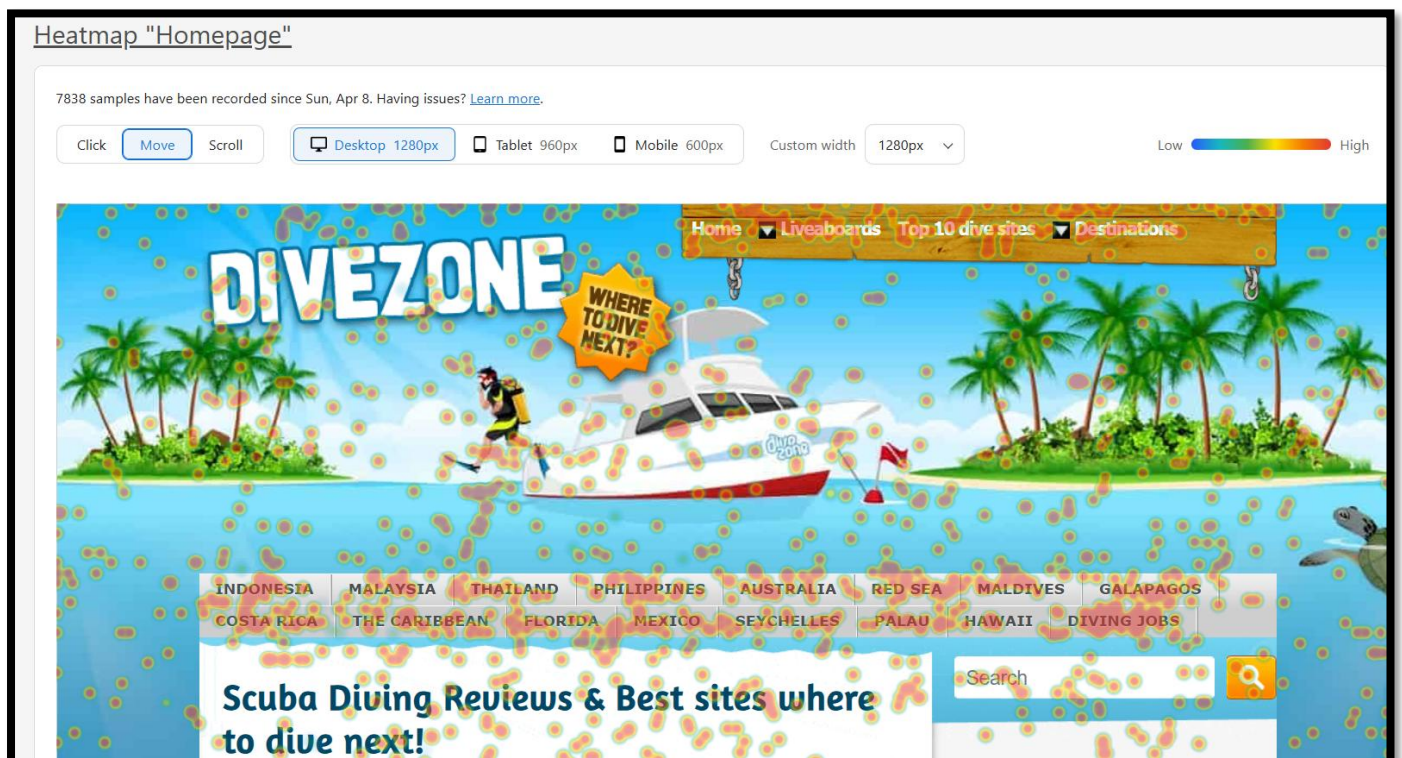
1. AI Insights:



2. Visitor Log:

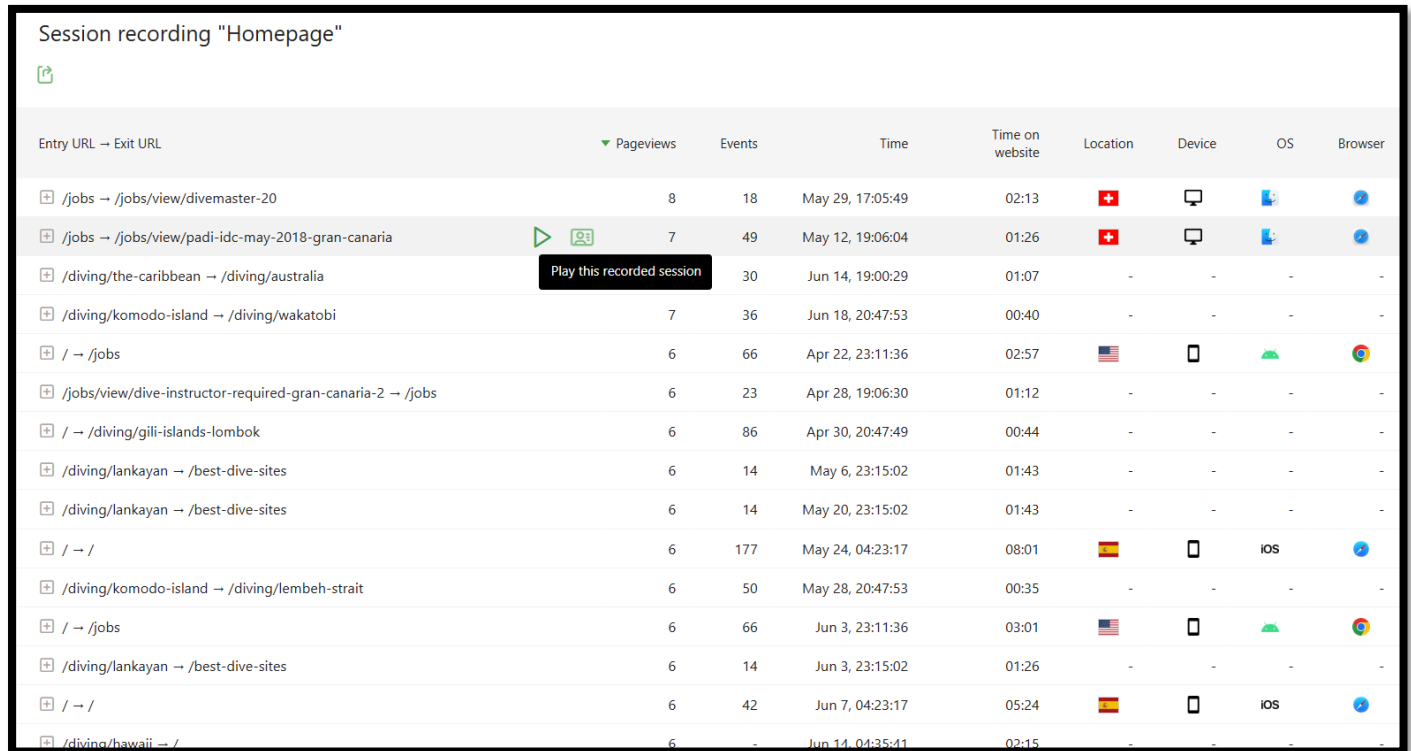


3. Heatmaps:



4. Session Recording

Session recording "Homepage"



Entry URL → Exit URL	Pageviews	Events	Time	Time on website	Location	Device	OS	Browser
/jobs → /jobs/view/divemaster-20	8	18	May 29, 17:05:49	02:13	🇨🇭	📱	📱	📱
/jobs → /jobs/view/padi-idc-may-2018-gran-canaria	7	49	May 12, 19:06:04	01:26	🇨🇭	📱	📱	📱
/diving/the-caribbean → /diving/australia	30	30	Jun 14, 19:00:29	01:07	-	-	-	-
/diving/komodo-island → /diving/wakatobi	7	36	Jun 18, 20:47:53	00:40	-	-	-	-
/ → /jobs	6	66	Apr 22, 23:11:36	02:57	🇺🇸	📱	📱	📱
/jobs/view/dive-instructor-required-gran-canaria-2 → /jobs	6	23	Apr 28, 19:06:30	01:12	-	-	-	-
/ → /diving/gili-islands-lombok	6	86	Apr 30, 20:47:49	00:44	-	-	-	-
/diving/lankayan → /best-dive-sites	6	14	May 6, 23:15:02	01:43	-	-	-	-
/diving/lankayan → /best-dive-sites	6	14	May 20, 23:15:02	01:43	-	-	-	-
/ → /	6	177	May 24, 04:23:17	08:01	🇪🇸	📱	iOS	📱
/diving/komodo-island → /diving/lembeh-strait	6	50	May 28, 20:47:53	00:35	-	-	-	-
/ → /jobs	6	66	Jun 3, 23:11:36	03:01	🇺🇸	📱	📱	📱
/diving/lankayan → /best-dive-sites	6	14	Jun 3, 23:15:02	01:26	-	-	-	-
/ → /	6	42	Jun 7, 04:23:17	05:24	🇪🇸	📱	iOS	📱
/diving/hawaii → /	6	-	Jun 14, 04:35:41	02:15	-	-	-	-

5. Goals and Conversions

Manage Goals

Only an Administrator or a user with Super User access can manage Goals for a given website.

Id	Goal Name	Description	Goal is triggered when	Revenue	Funnel	Attribution
4	New Job Application		Visit a given URL (page or group of pages) Pattern contains: /jobs/apply/	\$1	✓	✓
5	View Submit Job		Visit a given URL (page or group of pages) Pattern contains: jobs/add	\$5	✓	✓
6	New Resume		Visit a given URL (page or group of pages) Pattern contains: resumes/register	\$2	✓	✓
7	Liveboard.com click		Click on a Link to an external website Pattern contains: liveboard.com	\$1	✓	✓
8	Agoda click		Click on a Link to an external website Pattern contains: agoda	-	-	✓
9	User Comments		Goal is manually triggered using the JavaScript API trackGoal()	-	-	✓
10	Newsletter Signup		Visit a given URL (page or group of pages) Pattern contains: /newsletter/thank-you	-	-	✓